

TEST OF MATHEMATICS FOR UNIVERSITY ADMISSION

PAPER 1: APPLICATIONS OF MATHEMATICAL KNOWLEDGE

Practice paper · Set 02

75 minutes

INSTRUCTIONS TO CANDIDATES

Please read these instructions carefully, but do not open the question paper until you are told that you may do so.

A separate answer sheet is provided for this paper. You require a soft pencil and an eraser. Please complete the answer sheet with your candidate number, centre number, date of birth and full name.

This paper contains **20 questions**. For each question, choose the **one** answer you consider correct and record your choice on the separate answer sheet. If you make a mistake, erase thoroughly and try again.

There are no penalties for incorrect responses, only marks for correct answers, so you should attempt all 20 questions. Each question is worth one mark.

You can use the question paper for rough working or notes, but no extra paper is allowed.

You must complete the answer sheet within the time limit.

Calculators and dictionaries are NOT permitted.

There is no formulae booklet for this test.

Please wait to be told you may begin before turning this page.

This is a practice paper generated from a question bank written to the TMUA content specification. It is not an official UAT-UK paper. Answers and worked solutions are in the accompanying solutions booklet.

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- 1** The simultaneous equations $y = x^2 + 2x + c$ and $y = 4x + 1$ have exactly one solution. Find the value of c .
- A** 3
- B** -1
- C** 2
- D** 1
- E** -2
- F** 0

- 2 Which of the following sets of information is NOT sufficient to prove that two triangles are congruent?
- A two sides and a non-included angle equal (SSA)
 - B two sides and the included angle equal (SAS)
 - C two angles and a corresponding side equal (ASA)
 - D right angle, hypotenuse and one other side equal (RHS)
 - E three sides equal (SSS)

- 3** A sequence has first term 2 and each subsequent term is obtained by squaring the previous term and subtracting 2. What is the 50th term?

A 2^{50}

B -2

C 14

D 4

E 0

F 2

4 Solve $\log_2 x + \log_4 x = 6$. (You may use $\log_4 x = \frac{1}{2}\log_2 x$.)

A $x = 8$

B $x = 2^6$

C $x = 16$

D $x = 64$

E $x = 4$

F $x = 32$

5 The graph of $y = a(x + b)^2 + c$ has vertex $(-2, 5)$ and passes through $(0, -3)$. Find a .

A -2

B 2

C -4

D -1

E 1

F $-\frac{1}{2}$

6 Find the number of solutions of $\tan^2 x = 3$ for $0 \leq x < 2\pi$.

A 1

B 4

C 6

D 3

E 8

F 2

7 The tangent to $y = x^2 - 4x + 1$ at the point where $x = 3$ meets the y-axis at

A $(0, -2)$

B $(0, 8)$

C $(0, 4)$

D $(0, 1)$

E $(0, -5)$

F $(0, -8)$

- 8** The parabola $y = x^2 - 4x + 3$ is reflected in the x -axis and then translated 2 units in the positive y -direction. What is the maximum value of the resulting function?

- A** 3
- B** 1
- C** -1
- D** -3
- E** 5
- F** 2

9 The minute hand of a clock is 12 cm long. How far does its tip travel in 25 minutes?

A 25π cm

B 5π cm

C 10π cm

D 12π cm

E $\frac{25\pi}{6}$ cm

F 20π cm

- 10** Two cards are drawn at random from a standard pack of 52 without replacement. Which of the following expressions is the probability that both are aces?

A $\frac{4}{52} + \frac{3}{51}$

B $\frac{52}{4}$

C $\frac{52}{1} \times \frac{3}{52}$

D $\frac{13}{4} \times \frac{13}{3}$

E $\frac{52}{4} \times \frac{51}{4}$

F $\frac{52}{52} \times \frac{51}{52}$

- 11** How many three-digit positive integers have digits that are all different and all odd?
- A** 120
 - B** 24
 - C** 125
 - D** 100
 - E** 30
 - F** 60

- 12** The graph of $y = f(x)$ is transformed by the sequence: reflect in the x -axis, then translate 2 units in the positive y -direction, then reflect in the y -axis. The resulting graph has equation

A $y = f(-x) - 2$

B $y = 2 - f(x)$

C $y = -f(-x + 2)$

D $y = 2 - f(-x)$

E $y = -f(x - 2)$

F $y = -f(-x) - 2$

13 For all θ where the expressions are defined, $(\sin \theta + \cos \theta)^2 + (\sin \theta - \cos \theta)^2$ equals

A $2\sin^2 \theta$

B $2 + 2\sin \theta \cos \theta$

C 0

D 2

E $2\cos^2 \theta$

F 1

- 14** The curves $y = x^2 - 4$ and $y = -x^2 + 2x + k$ intersect at exactly two points. Find the complete set of values of k .

- A** $k > \frac{9}{2}$
B $k < -\frac{9}{2}$
C $k > -\frac{9}{2}$
D $k \neq -\frac{9}{2}$
E $k > -4$
F all real k

- 15** The second term of a geometric sequence is 6 and the fifth term is 48. Find the sum of the first 6 terms.
- A** 93
- B** 186
- C** 192
- D** 381
- E** 126
- F** 189

16 Which of the following is a factor of $x^4 - 5x^2 + 4$?

A $x - 3$

B $x^2 + 1$

C $x - 4$

D $x + 4$

E $x + 2$

F $x^2 + 4$

17 The area between the curve $y = x^2$ and the line $y = k$ (where $k > 0$) is $\frac{32}{3}$. Find k .

A 8

B 3

C $\sqrt{8}$

D 4

E 2

F 16

18 The number of integers n satisfying $2^n < 1000 < 2^{n+3}$ is

A 3

B 4

C 0

D 1

E 10

F 2

19 Find the sum of all real values of x satisfying $(x^2 - 3x)^2 - 2(x^2 - 3x) - 8 = 0$.

A 0

B 9

C 3

D -3

E 6

F 2

20 The curves $y = 2^x$ and $y = x^2$ intersect at how many points?

A 3

B 0

C 2

D 4

E 1

END OF TEST